CAPSTONE PROJECT

INTERVIEW TRAINER AGENT

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Power distribution systems are prone to various electrical faults that can disrupt operations, damage infrastructure, and threaten safety. These faults—such as line-to-ground, line-to-line, and three-phase short circuits—must be identified quickly and accurately to ensure grid stability. Traditional fault detection methods are often slow or manually intensive. There is a critical need for intelligent systems that can analyze electrical measurement data (voltage, current phasors) in real-time to detect and classify faults efficiently.

PROPOSED SOLUTION

The proposed system aims to address the challenge of detecting and classifying different types of faults in a power distribution system. This involves leveraging data analytics and machine learning techniques to identify and classify fault conditions accurately. The solution will consist of the following components:

Data Collection:
* Gather historical electrical measurement data, including voltage and current phasors under various operating conditions.
* Utilize real-time data sources from Intelligent Electronic Devices (IEDs), Phasor Measurement Units (PMUs), and SCADA systems to enhance detection accuracy.

Data Preprocessing:
* Clean and preprocess the collected data to handle missing values, noise, and inconsistencies.
* Perform feature engineering to extract relevant features (e.g., symmetrical components, phase angles, frequency variations) that influence fault detection.

Machine Learning Algorithm:
* Implement a machine learning algorithm, such as a classification model (e.g., Decision Tree, Random Forest, SVM, or Neural Network), to detect and classify faults based on electrical patterns.
* Consider incorporating additional features like load levels, system configuration, and protection relay status to improve classification accuracy.

Deployment:
* Develop a user-friendly interface or dashboard that provides real-time detection and classification of faults with visual alerts.
* Deploy the solution on a scalable and secure platform, considering factors like latency, reliability, and integration with existing grid infrastructure.

Evaluation
* Assess the model's performance using appropriate metrics such as Accuracy, Precision, Recall, F1-score, or Confusion Matrix.
* Fine-tune the model based on operational feedback and continuous monitoring of detection accuracy.

SYSTEM APPROACH

1. Upload historical electrical measurement data (voltage, current, etc.) to IBM Watsonx.ai with a labeled target column indicating the fault type.
2. Clean the data by handling missing values and outliers, and perform feature engineering to extract useful fault-related indicators.
3. Use Watsonx.ai AutoAI to automate model training by setting the prediction target as the fault type and selecting classification as the task.
4. Evaluate the model's performance using metrics like accuracy, precision, recall, F1-score, and analyze misclassifications.
5. Deploy the best-performing model in Watsonx.ai to create an API for real-time fault detection and classification.
6. Monitor the deployed model using Watson OpenScale, retrain periodically with new data, and refine the system for better reliability.

ALGORITHM & DEPLOYMENT

Algorithm Selection:

The proposed system uses IBM's AutoAI with classical machine learning algorithms such as **Random Forest**, **Gradient Boosting**, and **Support Vector Machines** available through Watsonx.ai. These models are selected for their high accuracy and robustness in classifying multi-class labels like different power system faults. Unlike deep learning models, these require less computation and are interpretable, making them suitable for real-time grid environments.

Data Input:

Electrical measurement data such as **voltage, current, frequency, and phase angles** from multiple nodes in the power distribution system are used as input features. Each record is labeled with a corresponding **fault type** (e.g., No Fault, LG, LL, LLG, LLLG), enabling supervised classification.

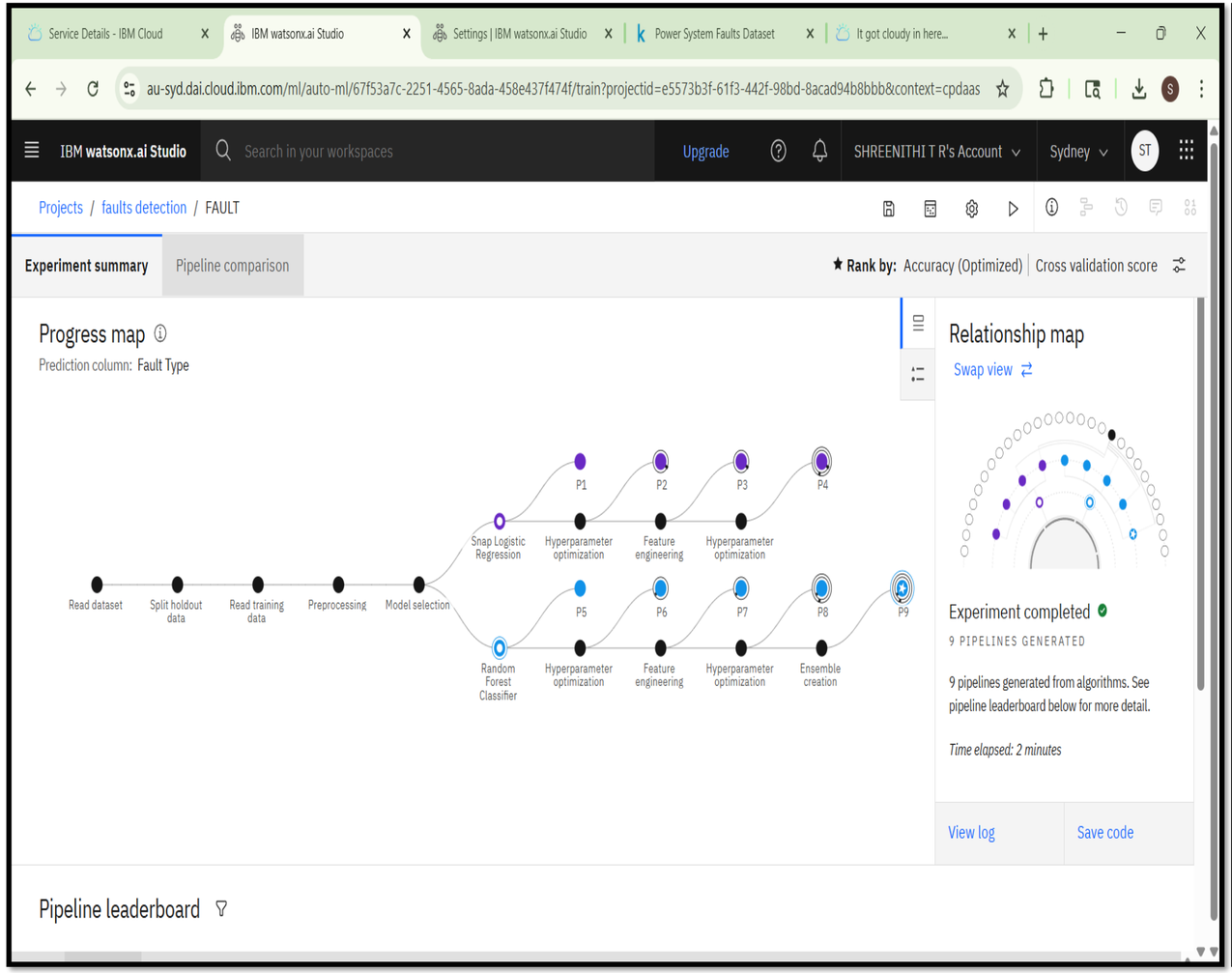
Training Process:

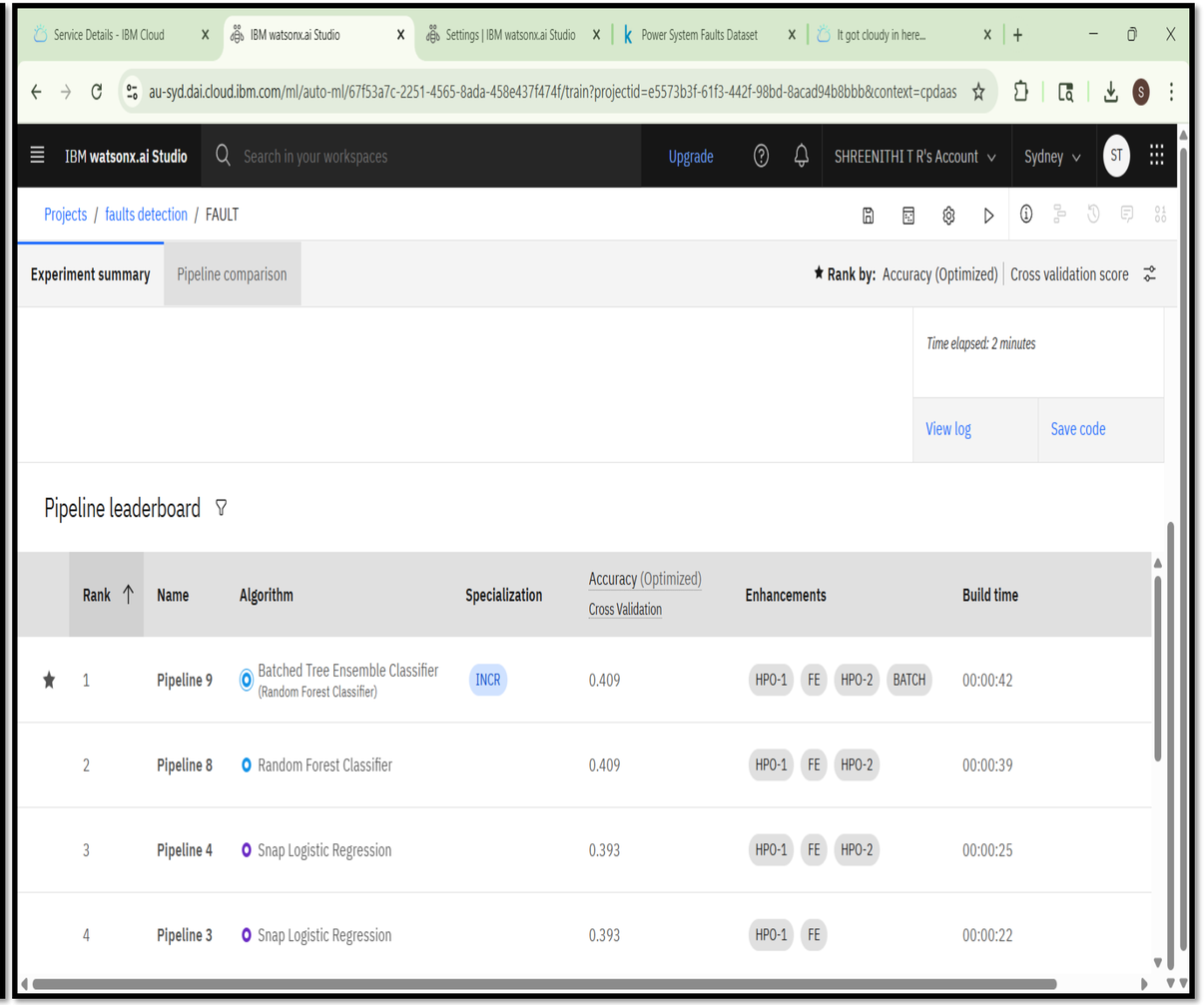
The uploaded dataset is cleaned, preprocessed, and fed into AutoAI. Watsonx.ai automatically selects relevant features, splits the dataset into training and test sets, and trains multiple models. It applies hyperparameter optimization and model comparison to select the best-performing fault classification model without the need for manual coding or advanced ML expertise.

Prediction Process:

Once deployed, the model receives real-time measurement data from the power grid and predicts the **fault type** instantly. The output is a classification result indicating whether the system is operating normally or under a specific fault condition. This enables timely fault isolation and helps maintain grid reliability.

RESULT





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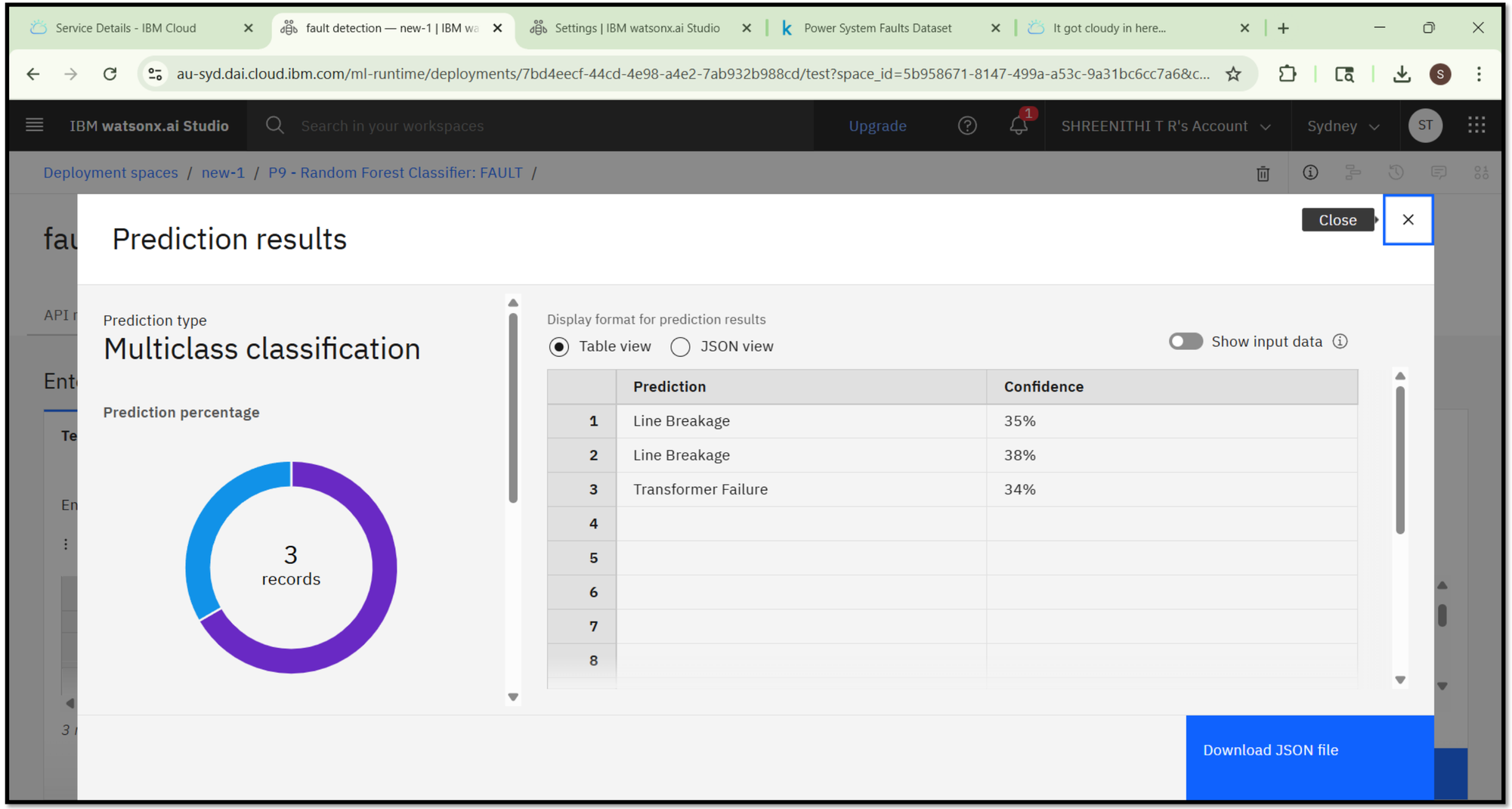
RESULT



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CONCLUSION

The implementation of Watsonx.ai for power system fault detection and classification demonstrates an efficient and scalable approach to automating fault analysis. By leveraging AutoAI, the system simplifies the machine learning workflow—handling data preprocessing, feature selection, model training, and optimization automatically. This not only accelerates model development but also ensures high accuracy and reliability in detecting and classifying various fault types. As a result, the system contributes significantly to improving the stability, responsiveness, and operational efficiency of modern power distribution networks.

FUTURE SCOPE

- In the future, the fault detection and classification system can be enhanced into a real-time monitoring and control solution integrated with smart grid infrastructure. The system can support **edge computing** for on-site processing, enabling faster response times during fault events.
- Additional capabilities such as **fault location estimation**, **severity analysis**, and **predictive maintenance alerts** can be incorporated to provide a more proactive and intelligent grid management solution. Integration with **IoT-based sensors** and **SCADA systems** will allow seamless data flow and automation.
- By leveraging **graph-based AI models**, **digital twins**, and **reinforcement learning**, the system can evolve into a self-learning, adaptive platform capable of optimizing grid performance and resilience under dynamic load and fault conditions.

REFERENCES

- IBM Watsonx.ai Documentation. https://www.ibm.com/docs/en/watsonx
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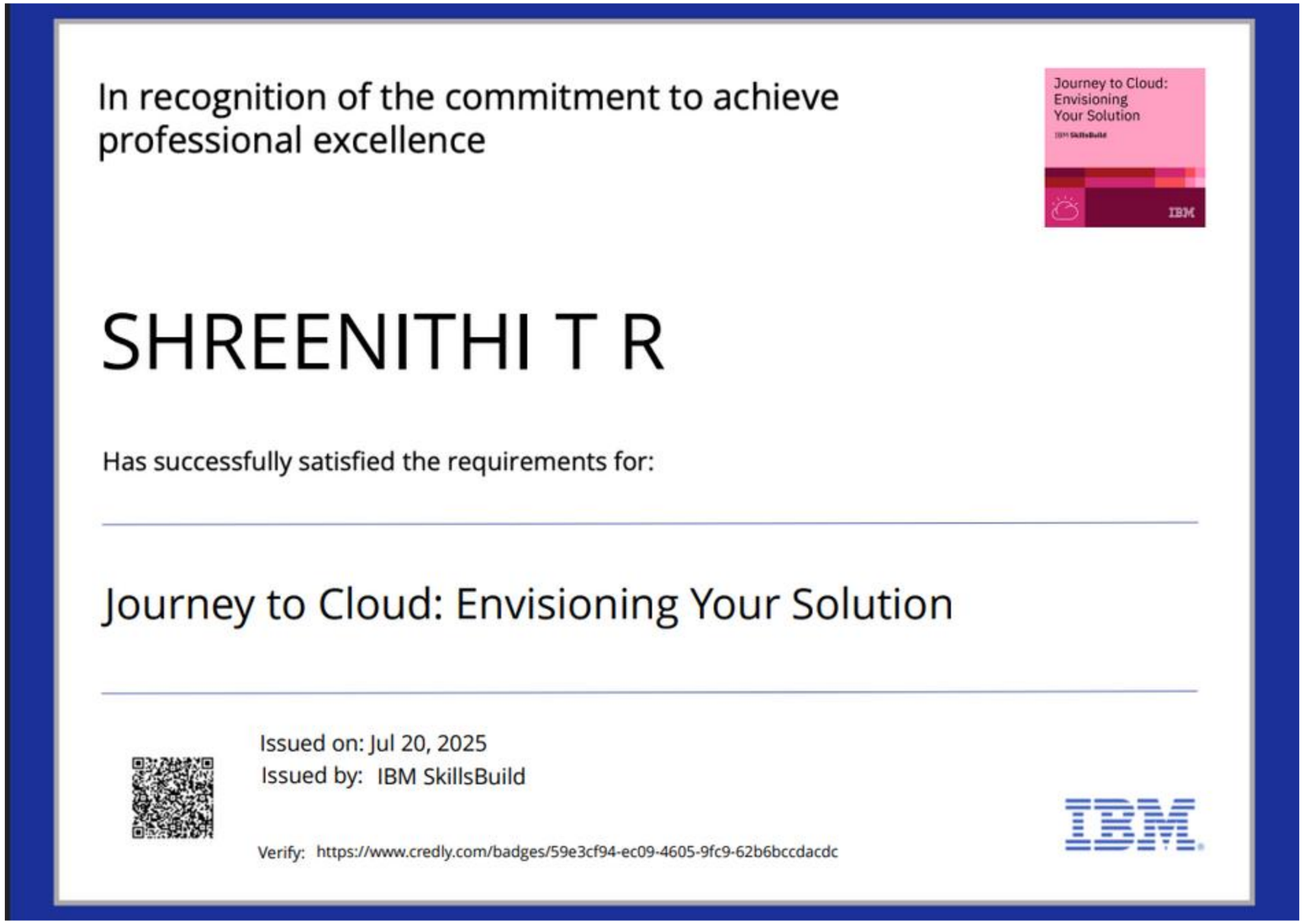

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